

Kunj Rathod

AI Research Engineer — Embodied AI, LLM Systems & Agents

kunj.rathod@utah.edu | +1 (385) 202-8879 | [linkedin/rathodkunj](https://www.linkedin.com/in/rathodkunj) | [github/rathodkunj2005](https://github.com/rathodkunj2005) | kunjrathod.com

EDUCATION

B.S. Computer Science — GPA 3.7/4.0, Dean's List

University of Utah

Aug 2023 – Dec 2026

Salt Lake City, UT

- Relevant: Machine Learning, Computer Vision, NLP, Distributed Systems, Interpretability of LLMs, Multimodal LLM Agents

RESEARCH

Spatial Memory for Embodied & Long-Horizon Web Agents

University of Utah

2026 – Present

- Designing retrieval-augmented memory architectures that unify spatial, temporal, episodic, and semantic memory for long-horizon robot navigation and manipulation (inspired by ReMemBR / NaVQA).
- Evaluating web agent benchmarks: OS Marathon, BrowseComp, Mind2Web; studying Web Explorer and Web Sailor V2 for trajectory-based multi-turn interaction and extended agentic reasoning.
- Studying SceneSmith (MIT CSAIL / Toyota Research) for procedural generation of simulation-ready environments from natural language for scalable robot policy evaluation.

Long-term Active Embodied QA (LA-EQA)

University of Utah | w/ Nikesh Subedi

Spring 2026

- Co-authored “Exploring Long-term Active Embodied QA in Simulated Indoor Environments.” Proposed **Video Mind Palace (VMP)**: replaced Robotic Mind Palace’s offline GPT-4o scene-graph pipeline with direct Qwen3-VL video queries, achieving **31–57% inference speedup** at 13–17% relative accuracy drop on the 100-question LA-EQA benchmark (single A100 GPU, Qwen3-VL 8B/30B/FP8).
- Identified three structural benchmark limitations and proposed future direction using SceneSmith + MuJoCo for interactive, multi-agent, long-horizon embodied QA evaluation.

Mechanistic Interpretability — Circuit Generalization in LLMs

University of Utah, advised by Ana Marasovic

2026 – Present

- Extracting sparse feature circuits from gemma-2-2b (GemmaScope transcoders) to predict generalization from default to counterfactual task distributions; built automated attribution graph pipelines (Pathways Discovery).
- Key negative result: suppressing over-refusal SAE features worsened over-refusal rates — publishable finding with implications for circuit-based behavior steering.

Multi-Agent Materials Discovery — STARS Lab

University of Utah | NASA, Microsoft, U.S. DoD collaboration

Aug 2025 – Feb 2026

Salt Lake City, UT

- Built graph-augmented multi-agent pipeline extracting material properties from 1,000+ papers for automated Ashby plot generation; developed constraint-based design region engine for aerospace materials selection.

EXPERIENCE

Software Engineer Intern — Azure Data

Microsoft

May 2026 – Present

Redmond, WA

- Building distributed data systems on Azure; full-stack engineering across the Azure Data ecosystem.

AI Services Intern — SUDO Program

University of Utah Health

Jan 2025 – Present

Salt Lake City, UT

- Built HIPAA-compliant LLM platform (AWS Bedrock Agents, Knowledge Bases, Guardrails) for 90+ hospital executives; p95 TTFT <200ms, 40% latency reduction, 1,000+ concurrent sessions via DynamoDB+S3 persistence.
- Shipped 6 full-stack features (React/TypeScript, Flask, Lambda) across 4 sprints; owned API design, schema, and infra.

AI Engineering Intern

CourtEasy.ai / Nugen

Nov 2024 – Apr 2025

Remote

- Hybrid RAG over 10M+ legal documents (dense + BM25 + reranking); +28% retrieval accuracy, -35% hallucination rate, 5,000+ daily queries.
- Benchmarked 8 LLM families on LegalBench, NyayaAnumana, IL-TUR; routing analysis saved \$50k+/yr projected inference cost.

AI Research Intern — BioGraphRAG

Garje Marathi Global

May 2024 – Aug 2024

Salt Lake City, UT

- Distributed GraphRAG over 1M+ biomedical entities (NebulaGraph); +40% factual accuracy, 3× traversal speedup to sub-500ms p95; ETL processing 2M+ entity updates/month. Article republished by NebulaGraph.

SELECTED PROJECTS

2024 - Present

- Metal-accelerated on-device LLM/VLM inference, Apple Vision OCR, CLIP embeddings, Graphiti-style temporal knowledge graph for multi-hop reasoning, local Whisper transcription, MCP server for agent interop. Zero cloud dependency.

2026

- End-to-end autonomous hiring: GitHub sourcing, live voice interviews (Twilio + transcript analysis), multi-dimensional scoring, Slack approval loop; replaced full human-in-loop recruiting workflow.

- **Exploring LA-EQA in Simulated Indoor Environments** — Rathod, Subedi. Class report, CS Advanced AI, U of Utah, Spring 2026.
- **BioGraphRAG** — co-authored; republished by NebulaGraph. [Substack](#), Oct 2024.
- **LLM Families on Legal Benchmarks** — internal analysis, CourtEasy.ai / Nugen, 2025.

Languages Python, C++, Java, TypeScript, Rust, SQL

Infra AWS Bedrock, Lambda, S3, DynamoDB, Docker, Kubernetes, PostgreSQL, NebulaGraph, ChromaDB, Pinecone

Kahlert Impact Prize (\$1,000) — Kahlert School of Computing, U of Utah, Mar 2026 · **Dean's List** 2024–25 · **Minute0 Hackathon Winner** Feb 2026